6/5/2025(E)

05 EXTRA

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: January 2025 –April 2025 Maximum Marks: 100 Examination: ESE Examination Duration: 3 Hrs.		
Programme code: 01 Programme: Computer Engineering	Class: SY	Semester: (SVU 2020) (V)
Name of the Constituent College: K. J. Somaiya School of Engineering	Name of the Department: Computer Engineering	
Course Code: 116U01C403	Name of the Course: Relational Database Management System	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Disadvantages of file systems w.r.t managing data	5
ii)	Give examples for representing Overlapping and Disjoint Class-subclasses relationship in EER diagram	5
iii)	Give Relational algebra queries to do aggregation operations	5
iv)	Explain Multilevel indexing with an example	5
v)	What are the different states of transaction, explain with diagram	5
vi)	Explain Deadlock handling during concurrency control	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain different types of users of DBMS	5
ii)	How Shared subclasses are represented in EER model, explain with an example	5
	OR	
Q2 A	Explain any 5 components of Database system architecture. Draw the diagram representing the components ofDatabase system architecture	10
Q 2 B	Any one	**10**
i)	Google wants to maintain a database of the placements and internships offered to colleges having details of packages offered, students, branches. Draw the EER diagram showing class- sub class, Participation, cardinality and relationships	10
ii)	List the steps used to map EER Model to Relational Model. Explain with example how composite, multivalued attributes, relationship with attribute from EER model are mapped to relational model	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Give SQL queries to do set operations with example	10
ii)	Give Relational algebra queries to do set operations with example	10
iii)	Explain Query Processing steps	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain 1NF, 2NF , 3NF, BCNF explain with example	10
ii)	What are the Desirable properties of decomposition, how to achieve it, Explain with example	10
iii)	How 2 phase protocol is used for concurrency control with example . Explain different types of 2 phase protocols.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Union and category	5
ii)	Relational algebra Group by having clause	5
iii)	TCL commands of SQL	5
iv)	Query Optimization	5
v)	Conflict serializable schedule	5
vi)	Different types of Locking Data	5